

Annex B

Compact RLA Topology for a Generalist Bryophyte

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Abstract

This annex specifies a fully computable Reticular Local Abstraction (RLA) topology for a generalist bryophyte. The topology is compact in the RLA/CRC sense: all relevant entities and rules are internal to the lattice (Internal Ontological Autonomy, IOA), predictions are produced without appeal to extra-reticular information (Epistemic Closure, EC), and every rule is algorithmic (Turing-Computability, TC). The package includes: (i) twenty abstraction levels (L01–L20); (ii) forty-two parameters (P01–P42) with units, literature ranges and operational reconduction rules; (iii) families of update equations and causal relations; (iv) threshold actions and rare-event injectors; (v) collapse / emergence metrics and composite indices; (vi) climate scenarios, initial state, halting / success and an adaptive mechanism. The bryophyte system is treated as an *epistemic* model: its levels, variables and equations encode disciplined scientific knowledge about the domain, without claiming to be an ultimate ontology of life. Synthetic trajectories generated by this topology are intended to be epistemically indistinguishable, under blind expert inspection, from empirical observations collected under analogous conditions.

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1 Introduction

Multi-level biological systems display macro-regularities that are not transparently reducible to micro-details (Anderson, 1972; Kitano, 2002; Godfrey-Smith, 2007). Bryophytes are a useful paradigm: from genomic regulation and chromatin state, through hormone and ROS dynamics, to tissue mechanics, demography and community interactions (Glime, 2007; Proctor, 2000, 2011).

Reticular Local Abstraction (RLA) represents such systems as lattices of levels linked by explicit transmission functions. Non-injective transmissions implement principled coarse-grainings and can support irreducible macro-properties; quasi-injective transmissions, when present, allow Turing-level undecidability to propagate upward in a controlled way.

In the terminology of Annex A, the structure constructed here is intended to be a compact reticulum satisfying Internal Ontological Autonomy (IOA), Epistemic Closure (EC) and Turing-Computability (TC), and therefore a concrete instance of Compact Reticular Computability (CRC) for a natural system. This annex provides a *compact* RLA topology for a generalist bryophyte: all explanatory work and all predictive power remain internal to the lattice, and the dynamics are fully Turing-computable. The lattice is designed as a computational reference case that can later be mapped to epistemic architectures (e.g. ECNNs) and to more abstract discussions on computability and emergence.

2 Methods

2.1 Abstraction levels L01–L20

The bryophyte is represented by twenty abstraction levels $L_{01}, \dots, L_{20}$. Each level L_i has a state domain $D(L_i)$ and an internal rule set $\Sigma(L_i)$. Levels are contiguous only when supported by recognised theories or measurements; ad hoc links are excluded.

Table 1 summarises the levels and their dominant processes.

Table 1: Abstraction levels for the bryophyte topology.

ID	Label	Description / dominant processes
L01	Genomic / Epigenetic	DNA sequence, transposable element activity, DNA methylation (5-mC), histone marks, stress-induced epimutations and chromatin changes (Joly & Matzke, 2018; Grob & Grossniklaus, 2020; Hirochika, 2000).
L02	Transcription / Translation	Transcript abundance (e.g. Rubisco), transcriptional noise, basic protein synthesis machinery (Perroud et al., 2018; Evans, 1989).
L03	Metabolic and Hormonal Regulation	Auxin, ABA, antioxidant enzymes and related metabolites; hormone-redox crosstalk and stress signalling (Cutler et al., 2010; Foyer & Noctor, 2009).
L04	Cellular Physiology	Reactive oxygen species (ROS), photosynthesis, turgor, osmotic and redox balance (Baker, 2004; Long et al., 2006).
L05	Tissues and Microstructures	Cell walls, lamina and costa, rhizoids, local biomechanics and micro-hydraulic properties (Baker & Doran, 2019; During & van Tooren, 2015).
L06	Morphogenesis / Architecture	Protonemal fronts, branching patterns, shoot layout and three-dimensional canopy structure.
L07	Reproductive Cycle	Spore dormancy, sporophyte nutrition and viability; link to fitness and long-term persistence (Hunt & Sparrow, 1974; Regmi et al., 2017).
L08	Biotic Interactions	Cyanobacterial colonisation, fungal symbiosis, microfauna grazing, and their net impact on nutrients and damage (Liu & Rousk, 2021; Pressel et al., 2015).
L09	Immediate Microenvironment	Near-canopy air temperature, light, wind and humidity; boundary layer processes (Garratt, 1994; Dow et al., 2021).
L10	Macro-climate Forcing	Seasonal cycle, inter-annual trends, heatwaves and droughts, parametrised on climate scenarios (IPCC, 2014, 2023).
L11	Stress Memory	Epigenetic and biochemical traces of past stress events (Pecinka & Mittelsten Scheid, 2011; Richards et al., 2016).
L12	Systemic Control / Homeostasis	Integrated control of vitality, stress indices and early-warning indicators (Scheffer et al., 2009; Dakos et al., 2012).

ID	Label	Description / dominant processes
L13	Soil and Substrate	Ion concentrations, pH, water retention and leaching (Peterson et al., 2015; Clymo, 1973).
L14	Endogenous Rhythms	Circadian oscillator and photoperiod coupling (Holm et al., 2010).
L15	Turnover / Senescence and PCD	Tissue ageing and apoptosis-like programmed cell death (Reape & McCabe, 2008; Petrov et al., 2015).
L16	Intercellular Signalling	Plasmodesmata permeability and peptide / auxin signalling (Sager & Lee, 2014; Tomoi et al., 2020).
L17	Geometric Patterning	Mesoscale order, phyllotaxis-like patterning, colony geometry (Grime, 1977; Waite & Sack, 2010).
L18	Tissue Mechanics	Elastic moduli, fracture thresholds, drought-induced mechanical changes (During & van Tooren, 2015).
L19	Evolution and Plasticity	Allele frequencies, genetic diversity, clonal structure, bottlenecks (Frankham, 2005; Szövényi et al., 2014; Miller et al., 2019).
L20	Systemic State and Meta-observability	Vitality, desiccation tolerance, cumulative stress indices and composite health metrics (Oliver et al., 2005; Proctor, 2011; Glime, 2007).

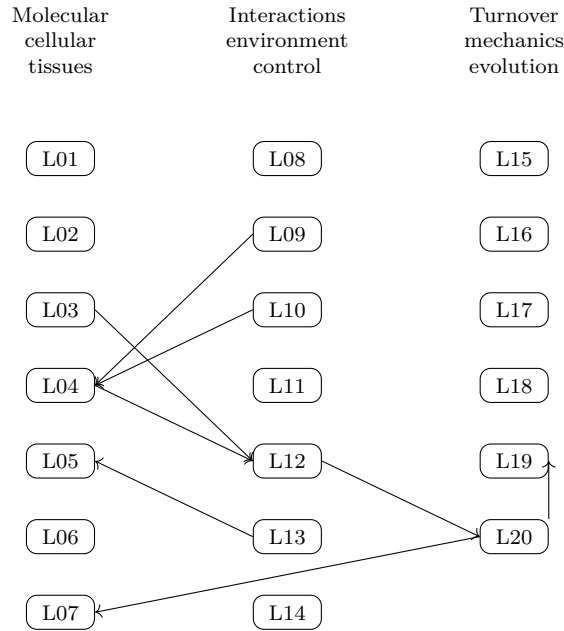


Figure 1: Schematic view of the twenty abstraction levels $L01$ – $L20$ and their dominant causal directions. Arrows highlight the main blocks (climate and substrate $\rightarrow$ physiology $\rightarrow$ systemic control and vitality $\rightarrow$ reproduction, demography and evolution).

2.2 Parameters P01–P42

The lattice state is encoded by forty-two scalar parameters $P01, \dots, P42$. Each parameter has a unit, a typical range extracted from the literature, and a reconduction rule mapping heterogeneous empirical values into a normalised domain.

Table 2 summarises names, units, indicative ranges and anchors.

Table 2: Core parameters (P01–P42): name, unit, indicative range, anchors.

ID	Name (unit)	Typical range	Literature anchors
P01	Rubisco expression (FPKM $\rightarrow$ a.u.)	0–1500 ($\approx$ 0.3–0.9 a.u.)	Perroud et al. (2018); Evans (1989).
P02	Auxin ($\mu\text{mol L}^{-1}$)	0.05–50	Suzuki et al. (2021).
P03	Antioxidant enzymes (U mg^{-1})	0.1–200	Usha et al. (2021); Foyer & Noctor (2009).
P04	ROS (H_2O_2 , $\mu\text{mol L}^{-1}$)	0.5–500	Hernandez et al. (2019); Baker (2004).
P05	Net photosynthesis ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$)	2–24	Long et al. (2006); Waite & Sack (2010).
P06	Leaf elastic modulus (MPa)	0.7–6.4	Baker & Doran (2019).
P07	PCD / apoptosis (%)	0–50	Ponce de León et al. (2012); Petrov et al. (2015).
P08	Circadian frequency (cycles day^{-1})	0.3–1.6	Holm et al. (2010).
P09	Fitness (a.u.)	0–1	Szövényi et al. (2014).
P10	Cumulative stress (a.u.)	0–10	Dakos et al. (2012); Proctor (2011).
P11	Vitality (%)	60–100 (healthy)	Oliver et al. (2005); Proctor (2011).
P12	Soil ion concentration (mM)	0.4–10	Britto & Kronzucker (2002); Peterson et al. (2015).
P13	Soil pH	3.0–8.0	Marschner (2012).
P14	Near-canopy temperature ($^{\circ}\text{C}$)	–4–48 (scenario-dependent)	Furness & Grime (1982); IPCC (2014).
P15	Incident light (lux)	8 000–120 000	Waite & Sack (2010); CIE (2020).
P16	Epimutational memory (5–mC, %)	4–8	Pecinka & Mittelsten Scheid (2011); Joly & Matzke (2018).

ID	Name (unit)	Typical range	Literature anchors
P17	Clonal stability (%)	0–100	Newton et al. (2007); Roberts et al. (2020).
P18	Epigenetic deviation (a.u.)	± 0.2	Bird et al. (1993).
P19	Transposon activity (events cell ⁻¹)	0–10	Hirochika (2000).
P20	Transcriptional noise (CV, %)	5–22	González et al. (2018).
P21	Intercellular flux (a.u.)	0.5–10	Tomoi et al. (2020); Sager & Lee (2014).
P22	Tissue differentiation (%)	0–100	Kierzkowski & Routier–Kierzkowska (2019).
P23	Spore dormancy (%)	0–100	Hunt & Sparrow (1974); Eklund et al. (2015).
P24	Cytokinin signalling (a.u.)	0–10	Takezawa et al. (2015); Thelander et al. (2018).
P25	Light competition index (0–1)	0–1	Chapin et al. (1993); Brodersen & Vogelmann (2008).
P26	Cyanobacterial mutualism (%)	5–60	Liu & Rousk (2021).
P27	Allele frequency (0–1)	0–1	Frankham (2005).
P28	Genetic diversity H_e (%)	20–80	Hughes et al. (2008); Miller et al. (2019).
P29	Normalised day of year (0–1)	0–1	ISO 8601 (ISO, 2019).
P30	Soil water content (m ³ m ⁻³)	0.15–0.40	Clark et al. (2016).
P31	Daily rainfall (mm day ⁻¹)	0–20	Rutter et al. (1971).
P32	Number of clones (n)	1–500	Newton et al. (2007).
P33	Clonal diversity (%)	0–90	Miller et al. (2019).
P34	Senescence (%)	0–80	Stewart et al. (2012).
P35	Surface water fraction (0–1)	0.1–1.0	Proctor (2011); Proctor & Brodribb (2021).
P36	Deep water fraction (0–1)	0.2–1.1 (clamped)	Dow et al. (2021).
P37	ABA (μmol L ⁻¹)	0.01–10	Cutler et al. (2010); Takezawa et al. (2015).
P38	Thermal trend (°C decade ⁻¹)	0–0.3	IPCC (2014, 2023); WMO (2023).
P39	Protonema development (a.u.)	0–1	Garcia et al. (2018).
P40	Sporophyte nutrition (a.u.)	0–1	Regmi et al. (2017).
P41	Sporophyte viability (a.u.)	0.70–1.0	Zhu & Ma (2017).

P42	Desiccation tolerance (relative water content, %)	0–100	Oliver et al. (2005); Griffiths et al. (2022).
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2.3 Operational recondution

Empirical measurements often come in heterogeneous units and with varying dynamic ranges. Each parameter is therefore mapped into a dimensionless simulation unit via a simple, explicit recondution rule $f_i : \mathbb{R} \rightarrow [0, 1]$ (or to a bounded subset of $\mathbb{R}$).

Examples:

- **Rubisco expression:**

$$P01_{\text{au}} = \frac{\text{FPKM}}{1500},$$

so that about 900 FPKM corresponds to 0.6 a.u., near empirical optima.

- **Soil water:**

$$P30 = \text{clamp}((\theta - 0.10)/0.30, 0, 1),$$

where θ is volumetric water content.

- **Desiccation tolerance:**

$$P42 = \frac{\text{RWC}}{100},$$

with RWC the relative water content in percent.

The clamping operator is defined as

$$\text{clamp}(x, a, b) := \max\{a, \min\{x, b\}\}, \quad (1)$$

and the indicator $\mathbf{1}\{E\}$ equals 1 if event E holds and 0 otherwise. These choices ensure boundedness and preserve the empirical ordering of states.

2.4 Update equations and causal relations

Time is discrete in days $t = 0, 1, 2, \dots$. The state at time t is the vector $\mathbf{P}(t) = (P01(t), \dots, P42(t))$. All equations are simple combinations of arithmetic operations, elementary functions (sine, exponential) and comparisons, ensuring Turing-computability.

2.4.1 Seasonal forcing and macro-climate

We model temperature, light and rainfall as bounded seasonal processes with an optional warming trend. Let $P29(t) \in [0, 1]$ be the normalised day of year.

$$P_{14}(t) = T_{\text{mean}} + A_T \sin(2\pi P_{29}(t)) + P_{38}(t), \quad (2)$$

$$P_{15}(t) = I_{\text{mean}} + A_I \sin(2\pi(P_{29}(t) - \varphi)), \quad (3)$$

$$P_{31}(t) = \max\{0, I_{\text{base}} + A_d \sin(2\pi(P_{29}(t) + \varphi_d))\}, \quad (4)$$

$$P_{29}(t+1) = (P_{29}(t) + 1/365) \bmod 1, \quad (5)$$

$$P_{38}(t+1) = P_{38}(t) + r_{\text{trend}}, \quad (6)$$

where T_{mean} , A_T , A_I , I_{mean} , I_{base} , A_d , φ , φ_d and r_{trend} are scenario parameters.

2.4.2 Water balance and ABA

Surface and deep water fractions and ABA follow:

$$P_{35}(t+1) = \text{clamp}\left(P_{35}(t) + k_s P_{31}(t) - k_e \max(0, P_{14}(t) - T_e), 0, 1\right), \quad (7)$$

$$P_{36}(t+1) = \text{clamp}\left(P_{36}(t) + k_d(P_{35}(t) - P_{36}(t)) - \lambda_{\text{H}_2\text{O}}, 0, 1\right), \quad (8)$$

$$P_{37}(t+1) = \text{clamp}\left(P_{37}(t)(1 - \gamma) + \beta \mathbf{1}\{P_{35}(t) < \theta\}, 0, P_{37,\text{max}}\right). \quad (9)$$

Here k_s is a surface recharge coefficient, k_e an evaporation parameter, k_d a diffusion coefficient from surface to depth, $\lambda_{\text{H}_2\text{O}}$ a slow loss term, and β, γ, θ control ABA dynamics.

2.4.3 ROS, antioxidants and photosynthesis

Baseline ROS are proportional to stress and supra-optimal temperature:

$$P_{04}(t) = \max\{0, a + bP_{10}(t) + c \max(0, P_{14}(t) - T_0)\}. \quad (10)$$

Antioxidant enzymes respond to ROS as:

$$P_{03}(t+1) = \max\left\{0, \min(P_{03,\text{max}}, P_{03}(t) + k_{\text{prod}} \max(0, P_{04}(t) - \text{ROS}_{\text{th}}) - k_{\text{decay}} P_{03}(t))\right\}. \quad (11)$$

Net photosynthesis is modulated by water, stress, temperature and CO_2 :

$$P_{05}(t) = P_{5,\text{base}}(0.6 + 0.4P_{35}(t))(1 - \kappa P_{10}(t)) \exp\left[-\frac{(P_{14}(t) - T_{\text{opt}})^2}{2\sigma^2}\right] \left[1 + \phi \frac{\text{CO}_2 - \text{CO}_{2,\text{ref}}}{100}\right], \quad (12)$$

where CO_2 and $\text{CO}_{2,\text{ref}}$ are ambient and reference CO_2 concentrations (ppm).

If ROS exceed a critical threshold, an oxidative penalty is applied:

$$\text{if } P_{04}(t) > \text{ROS}_{\text{crit}} : \quad P_{05}(t) \leftarrow \max \left\{ 0, P_{05}(t) \left(1 - \gamma \frac{P_{04}(t) - \text{ROS}_{\text{crit}}}{\text{ROS}_{\text{max}} - \text{ROS}_{\text{crit}}} \right) \right\}. \quad (13)$$

Alternative light / nutrient formulations (e.g. Michaelis–Menten or saturating light response) can replace (12) without changing the topology.

2.4.4 Hormones, stress, vitality and epigenetics

Intercellular flux and differentiation depend on auxin, antioxidants and senescence:

$$P_{21}(t) = \text{clamp} \left((\alpha P_{02}(t) + \beta P_{03}(t)) (1 - P_{34}(t)/M) \frac{P_{11}(t)}{100}, 0, P_{21,\text{max}} \right), \quad (14)$$

$$P_{22}(t) = \text{clamp} \left(k P_{21}(t) (1 - P_{34}(t)/N), 0, P_{22,\text{max}} \right). \quad (15)$$

Cytokinin signal is approximated as:

$$P_{24}(t) = \text{clamp} \left(\alpha_1 P_{02}(t) + \alpha_2 P_{14}(t), 0, P_{24,\text{max}} \right). \quad (16)$$

Cumulative stress P_{10} integrates ROS, water shortage and heat:

$$P_{10}(t+1) = \text{clamp} \left(P_{10}(t) + k_p \frac{P_{04}(t)}{\text{ROS}_{\text{max}}} + k_h (1 - P_{35}(t)) + k_t \frac{\max(0, P_{14}(t) - T_t)}{T_t} - k_{d,\text{rel}} P_{10}(t), 0, 10 \right), \quad (17)$$

where $k_{d,\text{rel}}$ is a relaxation coefficient (distinct from k_d in Eq. (8)).

Vitality depends on fitness, stress and temperature:

$$P_{11}(t) = \text{clamp} \left(\frac{100}{1 + \exp[-m(P_{09}(t) - \theta)]} - n P_{10}(t) - \max(0, P_{14}(t) - T_s), 0, 100 \right). \quad (18)$$

Epigenetic deviation is driven by antioxidant capacity:

$$P_{18}(t) = \text{clamp} \left(1 - \rho(100 - P_{03}(t))/100, -1, 1 \right). \quad (19)$$

2.4.5 Nutrients and pH

Soil ions and pH shape nutrient availability:

$$P_{12}(t+1) = \text{clamp} \left((m + n \sin(2\pi(P_{29}(t) + \varphi_n))) \max\{0.5, 1 - r|7 - P_{13}(t)|\}, P_{12,\text{min}}, P_{12,\text{max}} \right), \quad (20)$$

and nutrients feed back on photosynthesis through:

$$P_{05}(t) \leftarrow P_{05}(t) + k_9 \frac{P_{12}(t)}{K_9 + P_{12}(t)}. \quad (21)$$

Equivalently, in scenarios where pH dominates nutrient control, nutrient availability can be written as a Gaussian window in pH:

$$P_{12}(t) = P_{12,\max} \exp\left(-\frac{(P_{13}(t) - pH_{\text{opt}})^2}{2\sigma_{pH}^2}\right), \quad (22)$$

with Eq. (22) used in place of the pH term in Eq. (20).

2.4.6 Reproduction, morphology and desiccation

Morphogenesis and reproduction are modelled with simple recurrences:

$$P_{39}(t+1) = \text{clamp}\left(P_{39}(t) + k_g P_{05}(t) - k_{g,\text{decay}} P_{39}(t), 0, 1\right), \quad (23)$$

$$P_{40}(t+1) = \text{clamp}\left(P_{40}(t) + k_n P_{22}(t) - k_{n,\text{cost}} P_{40}(t), 0, 1\right), \quad (24)$$

$$P_{41}(t+1) = \text{clamp}\left(P_{41}(t) + k_v P_{40}(t) - k_{v,\text{sen}} P_{34}(t), 0.7, 1.0\right), \quad (25)$$

$$P_{42}(t+1) = \text{clamp}\left(P_{42}(t) + k'_d P_{16}(t) - k'_{d,\text{loss}}(1 - P_{35}(t)), 0, 1\right). \quad (26)$$

2.4.7 Causal relations

Equations (2)–(26) define a causal graph G across all parameters. Seasonal forcing drives temperature, light and rainfall; these drive water and ABA; water, stress and temperature shape ROS, antioxidants and photosynthesis; nutrients and pH modulate photosynthesis; hormones, stress and senescence determine vitality and reproduction; clonal structure feeds back into composite metrics. All maps are internal to the levels of Section 2.1.

2.5 Threshold actions and rare events

Threshold actions implement discrete jumps when guard conditions become true. They represent well-documented biological responses and remain local to the lattice. Table 3 provides a representative subset.

Table 3: Selected threshold actions.

ID	Guard (if true)	Immediate effect	Anchor
A01	$P_{04} > 120 \mu\text{mol L}^{-1}$	$P_{05} \leftarrow P_{05} - 3$ (oxidative damage to PSII)	Baker (2004).
A02	$P_{10} > 7$	$P_{07} \leftarrow P_{07} + 10$ (PCD increase)	Petrov et al. (2015).
A03	$P_{14} > 30^\circ\text{C}$	$P_{03} \leftarrow P_{03} + 5$ (heat-induced antioxidants)	Bitá & Gerats (2013).
A04	$P_{35} < 0.20$	$P_{06} \leftarrow P_{06} - 0.5$ (dehydration stiffening)	Proctor (2011).

ID	Guard (if true)	Immediate effect	Anchor
A06	$P_{15} > 60\,000 \text{ lux}$	$P_{16} \leftarrow P_{16} + 3$ (UV-linked methylation)	Boyko & Kovalchuk (2010).
A14	$P_{15} \in [40\,000, 70\,000] \text{ lux}$	$P_{11} \leftarrow P_{11} + 10$ (optimal light)	Waite & Sack (2010).
A18	$P_{04} > 120 \mu\text{mol L}^{-1}$	$P_{19} \leftarrow P_{19} + 3$ (ROS $\rightarrow$ TE activation)	Hirochika (2000).
A22	$P_{25} > 0.90$	$P_{32} \leftarrow P_{32} - 5$ (shade eliminates clones)	Chapin et al. (1993).
A29	$P_{34} > 60\%$	$P_{11} \leftarrow P_{11} - 10$ (senescence reduces vitality)	Stewart et al. (2012).
A29b	$50 < P_{04} < 120 \mu\text{mol L}^{-1}$	$P_{05} \leftarrow P_{05} + 2$ (ROS hormesis)	Mittler (2017).
A30	$P_{34} > 70\%$	$P_{07} \leftarrow P_{07} + 5$ (senescence $\rightarrow$ PCD)	Reape & McCabe (2008).
A35	$P_{14} < 15^\circ\text{C} \wedge P_{35} > 0.5$	$P_{10} \leftarrow P_{10} - 2$ (cool-humid relief)	Proctor (2011).
A_DES_0P ₄₂	$P_{42} < 0.20 \wedge P_{35} < 0.20$	$P_{05} \leftarrow P_{05} - 10$ (extreme desiccation stop)	Oliver et al. (2005).

Rare events inject low-probability shocks with internal effects only:

- *Pathogen*: daily probability $p_{\text{inf}} = 0.002$, $P_{04}+ = 20$, $P_{10}+ = 1$.
- *Fire*: $p_{\text{fire}} = 0.0005$, $P_{11}- = 30$, $P_{32}- = 10$.
- *Flood*: $p_{\text{flood}} = 0.001$, $P_{35}, P_{36} \mapsto 1.2$ (clamped to 1).

No new ontological entities are introduced; only existing state variables are perturbed.

2.6 Collapse, emergence and incoherence metrics

Metrics are computed on simulated trajectories $\{P_i(t)\}_{t=0}^T$. Standard statistics are used: $\text{var}(X)$ is the sample variance of a time series X , $\text{sd}(X)$ its standard deviation, and $\rho_1(X)$ the lag-1 autocorrelation. Each metric has (i) a target variable or composite; (ii) a statistic of its time series (mean, minimum, variance, slope, kurtosis, ratio); (iii) a threshold; and (iv) a biological interpretation.

Representative metrics:

- **Collapse metrics:**

$$M_{P_{11}, \text{crash}} := \mathbf{1}\{\min_t P_{11}(t) < 40\},$$

$$M_{P_{34}, \text{crit}} := \mathbf{1}\{\max_t P_{34}(t) > 90\},$$

$$M_{P_{05}, \text{collapse}} := \mathbf{1}\{\overline{P_{05}} < 5\},$$

corresponding respectively to vitality crash, irreversible senescence and near-zero photosynthesis.

- **Emergence metrics:**

$$\begin{aligned} M_{P10,\text{var}} &:= \mathbf{1}\{\text{var}(P_{10}) > \sigma_{P10}^2\}, \\ M_{P11,\text{AC1}} &:= \mathbf{1}\{\rho_1(P_{11}) > \rho_{\text{crit}}\}, \\ M_{P26,\text{vol}} &:= \mathbf{1}\{\text{sd}(P_{26}) > \sigma_{P26}\}, \end{aligned}$$

where σ_{P10} and σ_{P26} are scenario-specific thresholds. These track critical slowing down, instability and regime shifts (Scheffer et al., 2009; Dakos et al., 2012).

- **Incoherence metrics:**

$$M_{\text{H2O},\text{ratio}} := \mathbf{1}\left\{\frac{\text{var}(P_{35})}{\text{var}(P_{36})} > R_{\text{crit}}\right\},$$

indicating an incoherent separation of surface and deep water dynamics when R_{crit} is exceeded.

Composite indices at L20 (e.g. integrated stress, multifactor fracture) combine several metrics into single diagnostic flags.

2.7 Reticular decomposition and transmissions

For RLA purposes, the parameter set $\{P01, \dots, P42\}$ is partitioned across levels as follows (dominant associations):

- L01: $P16, P18, P19$ (epigenetic memory and TE activity).
- L02: $P01, P20$ (Rubisco transcription, transcriptional noise).
- L03: $P02, P03, P24, P37$ (auxin, antioxidants, cytokinin, ABA).
- L04: $P04, P05$ (ROS and photosynthesis).
- L05: $P06$ (tissue mechanical properties).
- L06: $P39$ (protonema / architectural growth).
- L07: $P23, P40, P41$ (spore dormancy and sporophyte status).
- L08: $P26$ (cyanobacterial mutualism).
- L09: $P14, P15$ (microclimate).
- L10: $P29, P31, P38$ (macro-climate forcings).
- L11: $P16, P18$ (stress memory, as in L01 but aggregated).
- L12: $P10, P11$ (cumulative stress and vitality).
- L13: $P12, P13, P30$ (substrate ions, pH, soil water).
- L14: $P08$ (circadian rhythm).

- L15: P_{07}, P_{34} (PCD, senescence).
- L16: P_{21}, P_{22} (intercellular flux and differentiation).
- L17: $P_{25}, P_{32}, P_{33}, P_{17}$ (competition index and clonal pattern).
- L18: P_{06} (mechanics, shared with L05 for local vs. global scales).
- L19: P_{09}, P_{27}, P_{28} (fitness and genetic diversity).
- L20: P_{35}, P_{36}, P_{42} plus composite metrics (systemic state).

Transmissions $\tau_{i \rightarrow j} : D(L_i) \rightarrow D(L_j)$ are *induced* by the update equations that carry information from one level to another. For example:

- $\tau_{04 \rightarrow 12}$: Eqs. (10) and (17) map cellular ROS (L04), together with water status (L20) and temperature (L09), into integrated stress P_{10} (L12). This transmission is strongly non-injective: many combinations of (P_{04}, P_{35}, P_{14}) lead to the same P_{10} .
- $\tau_{12 \rightarrow 20}$: Eq. (18) maps fitness (L19), stress (L12) and temperature (L09) into vitality P_{11} (L12) and, via metrics in Section 2.6, into composite systemic indicators at L20. This is also non-injective and collapse-inducing.
- $\tau_{13 \rightarrow 04}$: Eqs. (20)–(21) map substrate ions and pH (L13) into photosynthesis P_{05} (L04). On restricted domains this can be quasi-injective (monotonic in P_{12}), but globally it is non-injective due to saturation.

These induced transmissions realise the RLA reticulum for the bryophyte: levels L_{01} – L_{20} form the nodes, and the right-hand sides of Eqs. (2)–(26) and the threshold actions define the edges (transmissions) with explicit non-injective collapses at key interfaces (e.g. $L_{04} \rightarrow L_{12}$, $L_{12} \rightarrow L_{20}$). This structure underpins the compactness and CRC properties discussed in Section 5.

3 Scenarios, initial state and adaptive mechanism

3.1 Initial state

A default initial state corresponds to a healthy, moderately stressed bryophyte under benign climate:

- $P_{04} \approx 40$ (moderate ROS), $P_{03} \approx 100$ (good antioxidant defence),
- $P_{35} = 0.60$, $P_{36} = 0.70$ (favourable water status),
- ~ 100 clones (P_{32}) with $P_{33} \approx 60$,
- $P_{09} \approx 0.7$ (fitness) and $P_{11} \approx 80$ (vitality),
- $P_{23} \approx 10$ (low dormancy),
- $P_{16} \approx 4.5$, $P_{18} \approx 0$, $P_{19} \approx 1$ (low-stress epigenetics).

3.2 Climate scenarios

Macro-climatic parameters in Eqs. (2)–(6) are tuned to a small set of scenarios: baseline, moderate warming and severe warming, all compatible with recent CMIP/IPCC assessments (IPCC, 2014, 2023; WMO, 2023). Scenarios differ in T_{mean} , A_T , r_{trend} , and in the frequency / intensity of heatwaves and droughts, represented by temporary perturbations to P_{14} and P_{31} .

3.3 Halting and success conditions

Simulations halt when any of the following conditions is met:

- $t \geq T_{\text{max}}$ (time horizon reached);
- $\min_t P_{11}(t) < 25$ (catastrophic vitality loss);
- $\max_t P_{34}(t) > 90$ (almost complete senescence).

A run is counted as successful if, at halting,

$$\overline{P_{11}} > 50, \quad \overline{P_{23}} < 5, \quad \overline{P_{41}} > 0.7,$$

corresponding to sustained vitality, low dormancy and high sporophyte viability.

3.4 Adaptive epigenetic buffering

To reflect long-term plasticity, stress modulates epigenetic parameters:

- If $P_{10}(t) > 5$, then

$$P_{16}(t+1) = P_{16}(t) + 2, \quad P_{18}(t+1) = P_{18}(t) + 0.05,$$

and the thermal ROS gain c in Eq. (10) is reduced by a factor $(1 - 0.1 P_{16}(t)/100)$ (stronger buffering).

- If $P_{10}(t) < 3$, then

$$P_{16}(t+1) = 0.99P_{16}(t), \quad P_{18}(t+1) = 0.99P_{18}(t),$$

modelling relaxation towards baseline.

These rules are still purely algorithmic and operate entirely within the lattice.

4 Minimal Turing-compatible loop

The entire topology is implemented by a simple daily loop:

1. Update macro-climate variables using Eqs. (2)–(6).
2. Apply water and ABA equations (7)–(9).

3. Update ROS, antioxidants and photosynthesis using Eqs. (10)–(13).
4. Update hormones, stress, vitality and epigenetics (Eqs. (14)–(19)).
5. Update nutrients, reproduction and desiccation tolerance (Eqs. (20)–(26)).
6. Fire threshold actions and sample rare events.
7. Compute metrics and composite indices (Section 2.6).
8. Check halting / success and log state.

Each step requires a finite number of arithmetic operations, calls to elementary functions and comparisons. Therefore, for any finite horizon T , there exists a Turing machine that computes $\mathbf{P}(0), \dots, \mathbf{P}(T)$.

5 Compactness and epistemic indistinguishability

5.1 Compact RLA/CRC properties

Let $L = \{L01, \dots, L20\}$ and $T = \langle L, \{f_{ij}\}, \Sigma \rangle$ be the bryophyte lattice, where f_{ij} are transmission functions between levels and Σ is the finite set of update rules (equations, actions, rare events, metrics).

Internal Ontological Autonomy (IOA). All state variables $P01$ – $P42$ live on levels in L ; all update rules in Σ manipulate only these variables. Exogenous drivers (climate forcings) are represented via internal parameters at $L09$ – $L10$. No rule refers to entities outside the lattice. Many f_{ij} are deliberately non-injective, collapsing microstates into aggregated macro-states (e.g. vitality, desiccation indices), thereby preventing external undecidability from leaking in.

Epistemic Closure (EC). For any proposition about the system (e.g. “vitality will stay above 50 % over the next 100 days under scenario S ”), truth within the model is decidable by iterated application of the internal prediction operator P_t induced by Σ . No external oracle, dataset or law is required. All observational predicates refer to variables at levels $L01$ – $L20$.

Turing–Computability (TC). Each element of Σ is a composition of Turing–computable functions on rationally representable reals. Thus, for any finite horizon $T \in \mathbb{N}$ and any initial state $\mathbf{P}(0)$, there exists a finite algorithm that computes $\mathbf{P}(1), \dots, \mathbf{P}(T)$. This suffices for Turing–computability for all practical purposes.

The bryophyte lattice is therefore a compact reticulum in the sense of Compact Reticular Computability (CRC).

5.2 Epistemic indistinguishability

Definition (epistemic indistinguishability). Let S be a fixed set of observables and let O be an observer restricted to S . A real trajectory τ^{real} and a simulated trajectory τ^{sim} are *epistemically indistinguishable for O at threshold η* if, given only the restricted outputs $\tau^{\text{real}}|_S$

and $\tau^{\text{sim}}|_S$, no decision procedure available to O can assign which trajectory is real with success probability strictly greater than η .

In the bryophyte setting, S might contain only a subset of variables (e.g. biomass proxies, fluorescence, coarse stress indices) observed at finite temporal resolution. If an expert observer O restricted to S cannot reliably distinguish real from synthetic trajectories above a chosen threshold η (e.g. $\eta = 0.7$), then, relative to S and η , the compact RLA topology is empirically adequate in the sense of this definition.

Under the RLA view, a compact topology can in principle produce synthetic time series of observables that are epistemically indistinguishable from empirical data, provided that:

1. the micro–foundation (levels, parameters, ranges) is consistent with current biological knowledge;
2. the observation protocol for the simulation matches that of real experiments (same variables, aggregation rules and noise);
3. the expert observer has no access to extra–lattice information about data provenance.

A blind test proceeds as follows:

- generate simulated trajectories of key observables (e.g. proxies of biomass, fluorescence, stress indices) under a chosen climate scenario;
- collect real trajectories from controlled microcosms under comparable conditions;
- mix and anonymise both sets and present them to a bryophyte expert;
- ask the expert to classify each series as “real” or “synthetic” and compare performance with chance.

If no statistically significant discrimination is possible, this does not prove ontological identity between simulation and reality. However, it supports the claim that the compact RLA topology captures sufficient structure of the system to reproduce its observable dynamics in the chosen domain of discourse. This constitutes a falsifiable prediction of the framework and an analogue of a Turing test for empirical vs. computed trajectories.

6 Discussion and limitations

The compact RLA topology for a generalist bryophyte illustrates how a non–trivial biological system can be represented as a finite, multi–level computational object with clear semantics and explicit closure properties.

Key points:

- The design is intentionally *epistemic*. Levels and parameters encode a discretised scientific understanding of the organism, not an ultimate ontology. Empirical ranges and rules are drawn from the literature but remain revisable.

- Non-injective transmissions are used as constructive design tools: they encode how scientists legitimately aggregate micro-details into macro-indices (e.g. vitality, stress, desiccation). Irreversible information loss at these interfaces explains how emergent properties can be both real (for the model) and computationally manageable.
- Collapse and emergence metrics provide a bridge from dynamical systems analysis (critical transitions, early-warning signals) to practical diagnostics for the bryophyte, and more generally, for complex systems with similar multi-level structure.

Limitations include the absence of explicit spatial heterogeneity, simplified treatment of species differences, and uncertainty in many parameter estimates. Nonetheless, the topology is rich enough to support stress tests, scenario analysis and epistemic experiments about the boundary between simulated and empirical realities.

Sensitivity (conceptual)

The present topology is not calibrated in a strict statistical sense, but its structure makes it possible to reason qualitatively about sensitivity and robustness. A few parameters and thresholds play a critical role:

- **Stress accumulation and relief (P10 and related gains).** Increasing the gains k_p, k_h, k_t in Eq. (17) makes cumulative stress grow faster for the same ROS, drought and heat patterns. Trajectories cross the collapse metrics $M_{P11, \text{crash}}$ and $M_{P34, \text{crit}}$ more often, and the lattice exhibits earlier and more frequent transitions to low-vitality regimes. Decreasing these gains has the opposite effect and extends the basin of attraction of high-vitality trajectories.
- **Climate forcing (parameters for P_{14} and P_{31}).** Raising T_{mean} , A_T or the warming trend r_{trend} in Eqs. (2)–(6) increases the frequency of excursions above the thermal thresholds in the stress and ROS equations. In RLA terms, transmissions from climate levels (L09–L10) to physiological levels (L03–L04) become more strongly non-injective: a larger fraction of distinct climate microstates collapses onto similar “overstressed” physiological configurations, reducing resilience.
- **Water status (P35, P36).** Modifying k_s, k_e, k_d in Eqs. (7)–(8) changes how rapidly surface and deep water respond to rainfall and heat. Larger k_e or smaller k_s sharpen drought episodes at L09 and L13 into more frequent activation of threshold actions such as `A_DES_0`, effectively strengthening the collapse from water dynamics to photosynthesis and vitality. Conversely, parameter choices that maintain P_{35} and P_{36} in intermediate ranges widen the set of micro-trajectories that remain in high-vitality macro-states.
- **Vitality non-linearity (P11).** The slope and threshold parameters in Eq. (18) control how sharply vitality responds to fitness and stress. A steeper sigmoidal function turns moderate increases in P_{10} into abrupt drops in P_{11} , making collapse metrics more sensitive and increasing the reticular irreducibility index: more of the lattice must be taken into account to anticipate tipping points.

- **Epigenetic buffering (P16, P18).** The rules in the adaptive mechanism section couple high stress (P_{10}) to increases in P16 and P18, which in turn reduce the thermal ROS gain in Eq. (10). Strengthening this feedback (larger increments in P16, P18) can delay or even prevent collapse for certain climate scenarios, effectively weakening the collapse of climate trajectories onto a small set of stressed physiological states. Weakening the feedback has the opposite effect and makes the lattice more fragile to small changes in forcing.

From an RLA/CRC perspective, these qualitative sensitivities can be read as statements about how the non-injective transmissions between climate, physiology and systemic levels change with parameters. Stronger non-injective collapse at critical interfaces (e.g. L09–L10 $\rightarrow$ L03–L04, L12 $\rightarrow$ L20) corresponds to reduced resilience and more frequent entry into collapse metrics; parameter regimes that maintain a richer variety of mesoscopic states at these interfaces correspond to wider basins of attraction for healthy regimes.

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